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Final Year Project Report

Yogi Amitkumar Patel
2536809

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Supervisor: Maimoona Sarif
CN6000

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Abstract

Acknowledgments

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2.1 Introduction

2.1.1 Clinical Background and the "Black Box" Challenge

In modern oncology, properly identifying and managing brain tumours continues to be a major issue. Due to its simplicity and excellent volumetric resolution, Magnetic Resonance Imaging (MRI) has become the highest standard for analysis (Menze *et al.*, 2015). However, there is a substantial barrier in the interpretation of this data. At the moment, the process of defining the boundaries of tumors is mostly done by hand. Subjectivity and labour intensity are introduced by this reliance on human interpretation, which may not be sustainable for scaling healthcare solutions (Iftikhar *et al.*, 2025). Automated technologies that can match or surpass human precision without the corresponding time restrictions are therefore desperately needed in medical applications.

The response to this clinical need has been the rapid adoption of Deep Learning (DL) technologies, particularly 3D Convolutional Neural Networks (CNNs) like the 3D U-Net. These models have demonstrated "superior performance" in handling volumetric data compared to traditional methods (Bhati, Neha and Amiruzzaman, 2024; Iftikhar *et al.*, 2025). However, the integration of these advanced models has introduced a new, critical issue: the **"black box"** problem.

While these models are highly accurate, their internal decision-making processes are opaque (Neri *et al.*, 2023). **This creates a paradox in medical AI**: as models become more complex and accurate, they often become less interpretable to the clinicians using them. In a high-stakes environment like neurosurgery, a lack of transparency creates a "trust gap" (Wen *et al.*, 2025). It is not enough for a model to simply output a segmentation mask; clinicians require the ability to verify why specific voxels were classified as tumour tissue. Therefore, the lack of interpretability is no longer just a technical issue, but a barrier to ethical and safe clinical deployment (Neri *et al.*, 2023).

2.1.2 Research Aims and Scope of Literature Review

To address this disconnect between model accuracy and clinical trust, this literature review aims to critically evaluate the intersection of 3D deep learning, Explainable AI (XAI), and immersive visualisation. While these fields are often studied in isolation, this review argues that a unified framework is necessary to fully illuminate the "black box" of 3D CNNs.

The review is structured to build this argument logically:

Section 2.2 establishes the technical capabilities of modern 3D CNN architectures for segmentation.

Section 2.3 critically analyses current XAI methods (such as Grad-CAM and LIME), evaluating their limitations when applied to complex volumetric data.

Section 2.4 explores the potential of immersive technologies (VR) to present these XAI outputs in a way that is intuitively understandable for clinicians.

Section 2.5 synthesises these findings to identify the specific research gap this project will address: the lack of an integrated, 3D-visualised explainability pipeline for brain tumour segmentation.

Section 2.6 explains the investigation's findings and explains how this initiative will fill the identified gap.

2.2 3D Deep Learning for Medical Brain Tumour Segmentation

2.2.1 Evaluation of 3D CNNs for Brain Tumour Segmentation

Over the past decade, the field of medical image analysis has experienced a significant paradigm shift, moving from **"reasoning-based"** systems that relied on manual feature engineering to **"learning-based"** deep learning (DL) approaches (Neri *et al.*, 2023). In the past, radiomics was used for the automated analysis of brain tumours. To train classifiers like Support Vector Machines (SVMs), domain specialists manually retrieved quantitative features including texture, intensity, and shape. However, this method was labour-intensive, prone to observer bias, and frequently failed to capture the intricate, non-linear spatial relationships present in volumetric data (Bhati, Neha and Amiruzzaman, 2024).

The introduction of Convolutional Neural Networks (CNNs) brought an end to this manual paradigm. CNNs prefer an end-to-end approach to extract features automatically by utilizing raw pixel data in contrast to classical machine learning. Initial implementation in neuro-oncology treated Magnetic Resonance Imaging (MRI) as a series of 2D slices. While this approach allowed taking pre-trained networks into consideration (such as VGG or ResNet), it has a significant drawback: it neglects the spatial context along the z-axis (depth).

In neuro-oncology, a tumor is an inherently volumetric entity; it defines the anatomical boundaries with continuation across the sagittal, coronal and axial planes. When a 3D volume is processed as a series of 2D slices it frequently results in **"discontinuous predictions"**, which produce uneven and anatomically impossible 3D models where a lesion is found in one slice but overlooked in the adjacent one (Iftikhar *et al.*, 2025). To overcome this problem, the field has moved towards 3D CNNs, which simultaneously conduct convolutions on the entire (x, y, z) volume using volumetric kernels (such as $3 \times 3 \times 3$). This volumetric approach allows the model to learn complex spatial hierarchies and inter-slice connections, which is essential for precise segmentation of brain tumors using large-scale datasets like BraTS.

2.2.2 Evolution of Volumetric Segmentation Architectures

To handle the computational complexity of 3D data while conducting segmentation that is pixel-perfect, the research community has standardized the **"encoder-decoder"** architecture.

Building upon the success of 2D U-Net, Çiçek et al. (Çiçek *et al.*, 2016) introduced the 3D U-Net architecture, a fully convolutional network for dense volumetric image segmentation. A U-shaped symmetric model consists of two distinct pathways:

1. **The Contracting Path (Encoder)**: This pathway significantly considers feature extraction. It consists of max-pooling layers after recursive blocks of 3D convolutions. As data is processed through the encoder, the spatial resolution decreases while the number of features increases. This path allows the model to capture high-level conceptual data context ("where is the tumour present"), such as differentiating tumour tissues from healthy ones (Çiçek *et al.*, 2016).
2. **The Expanding Path (Decoder)**: To regenerate the segmentation map, the spatial resolution that was lost during encoding needs to be restored. This path reconstructs the feature maps back to their initial input dimensions using up-convolution (transposed convolution) layers, also providing precise localisation ("where is the tumour").

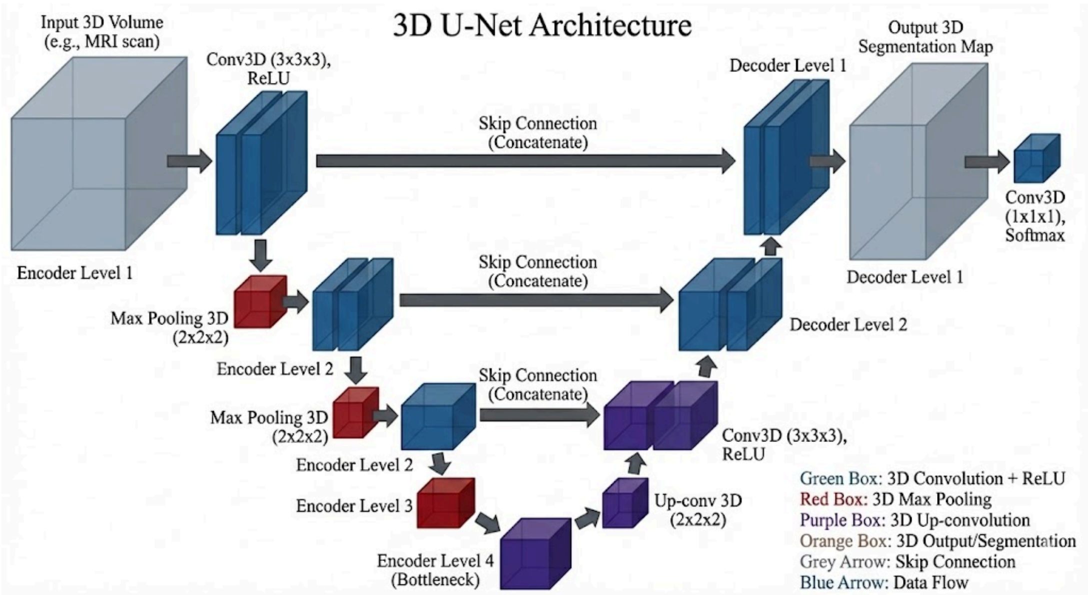


Figure 1: 3D U-Net Architecture

A critical innovation of the 3D U-Net introduced long-range skip connections. Deep neural networks often face difficulty acquiring fine-grained spatial information after applying multiple pooling operations. The 3D U-Net addresses this problem by concatenating high-resolution feature maps straight from the encoder to the appropriate decoder layers. According to Çiçek et al. (Çiçek *et al.*, 2016), this mechanism allows the network to incorporate deep semantic features with shallow details, with precise boundary delineation even when trained with sparse annotations.

Despite its success, the standard U-Net struggles with small lesions. Medical images often contain irrelevant background information such as non-considerable tissue in the brain, while the standard U-Net treats all the pixels in a feature map equally. To overcome this issue, researchers have integrated "attention mechanisms" into the U-Net architecture. As explained by Natekar et al. (Natekar, Kori and Krishnamurthi, 2020), the Attention U-Net incorporates **"attention gates"** into the skip connections.

These gates utilise the coarser signal from the gating vector (the deep layer) to filter the activations from the input signal (the shallow layer) before they are merged. This enhances the activations in the Region of Interest (ROI) and suppresses the irrelevant background regions using mathematical equations. This "soft-attention" mechanism improves the model's sensitivity to small, irregular lesions such as glioma subregions without requiring additional guidance or complex cropping preprocessing (Natekar, Kori and Krishnamurthi, 2020).

Parallel to the U-Net, Milletari et al. (Milletari, Navab and Ahmadi, 2016) proposed the V-Net, an optimised model specifically for volumetric medical data. V-Net is completely distinct from other models, incorporating **residual connections** (short-skip connections) blocks. This model resembles the ResNet philosophy, adding the input of a block to its output, allowing gradients to flow more smoothly during backpropagation. Compared to convolutional U-Net, this reduces the vanishing gradient issue and makes it easier to train much deeper 3D networks (Milletari, Navab and Ahmadi, 2016).

Furthermore, the V-Net research has presented a solution to one of the most enduring challenges in medical segmentation: class imbalance. In a dataset, tumour occupies less than 1% of the total volumetric pixels while the rest of the area is unaffected and healthy background. Standard objective functions like cross-entropy fail in this scenario, resulting in a model achieving accuracy of 99% by predicting "background irrelevant data" for every voxel. To overcome this, Milletari et al. (Milletari, Navab and Ahmadi, 2016) introduced the **Dice coefficient**, which optimises the model by considering the gap between the ground truth and the expected segmentation:

$$D = \frac{2\sum_i^N p_i g_i}{\sum_i^N p_i + \sum_i^N g_i}$$

Where the sums run through the N voxels, of the predicted binary volume $p_i \in P$ and the ground truth binary volume $g_i \in G$. This function can be improved by rewriting it in a differentiable form. As a result, it can be optimised with gradient descent. It can be computed with respect to each predicted voxel, allowing the network to improve the overlap between the predictions and ground truth (Milletari, Navab and Ahmadi, 2016).

The gradient:

$$\frac{\partial D}{\partial p_j} = 2\left[\frac{g_j}{\sum_i p_i^2 + \sum_i g_i^2} - \frac{2p_j\left(\sum_i p_i g_i\right)}{\left(\sum_i p_i^2 + \sum_i g_i^2\right)^2}\right]$$

2.3 Explainable AI (XAI) for Explainability in Medicine

2.3.1 The Need for Explainability in Clinical Practice

The integration of Artificial Intelligence (AI) into medical practice, especially in high-stakes domains like neuro-oncology, has been made more difficult by the opacity of modern Deep Learning (DL) models. Convolutional Neural Networks have demonstrated superior performance in segmentation-like tasks, yet their complex, non-linear decision-making remains inaccessible. This "black box" nature has become a barrier to model adoption, often referred to as the **"trust gap"** (Wen *et al.*, 2025).

Just model prediction is insufficient in clinical practice; it must incorporate justification that follows medical knowledge. As highlighted by Neri et al. (Neri *et al.*, 2023), the ethical and legal framework governing medicine, such as the "right to explanation" under GDPR, demands that automated decisions be transparent and contestable. Clinicians must be able to identify the reason behind the model classification of a specific region as tumour-present or not to ensure patient safety (Iftikhar *et al.*, 2025).

Furthermore, XAI is not just a compliance tool but also a critical mechanism for model debugging and knowledge discovery. Researchers can detect "clever Hans" events, in which a model learns misleading connections (e.g., forecasting tumour based on a specific hospital watermark rather than anatomical pathology) (Hou *et al.*, 2024). When it comes to brain tumour segmentation, XAI refers to the validation of network focus on biologically relevant sub-regions such as the necrotic core or enhancing tumour (Natekar, Kori and Krishnamurthi, 2020).

2.3.2 Review of Voxel-Based XAI Techniques

The literature categorizes XAI approaches according to their scope and timing relative to model training. A primary distinction is made between ante-hoc (or "intrinsic") and post-hoc methods.

- **Ante-hoc:** Methods designed to be interpretable by their nature. For example, decision trees or linear regression models, where the relationships between them are explicit (Agrawal *et al.*, 2025). By integrating interpretable prototype blocks directly into the network architecture, recent advancements in "Self-Explainable AI (S-XAI)" plan to introduce transparency in deep learning (Hou *et al.*, 2024).
- **Post-hoc:** Methods designed for explainability after the model has been trained. They treat a model as a "black box" and try to improve and approximate its behaviour based on the given inputs and outputs. This method is preferred as standard for analysing medical imaging models like the 3D U-Net architecture to perform segmentation (Bhati, Neha and Amiruzzaman, 2024).

Further XAI methods are divided by their scope:

- **Global explanations:** Methods that explain the overall logic of the model across the entire dataset, such as feature importance in a model.

- **Local explanations:** Methods that explain the decision-making process of the model for specific inputs, such as the reason behind classifying a voxel region as tumour. Local explanation is important for clinical decision-making as they offer the patient-specific rationale for diagnosis and therapy planning (Iftikhar *et al.*, 2025).

For 3D medical imaging, the most common XAI techniques are those that can generate saliency maps, spatial representations of "feature importance". The three most prominent approaches are **Grad-CAM**, **LIME** and **SHAP**.

Gradient-Weighted Class Activation Mapping (Grad-CAM) has been recognised as the "golden standard" for visualising CNN decisions in medical imaging (Bhati, Neha and Amiruzzaman, 2024). Grad-CAM is model-agnostic and can be applied to any CNN model without retraining, compared to earlier methods that required architectural changes.

Mechanism: Grad-CAM utilises the gradient of the target concept (e.g., tumour) to produce a saliency map, which highlights the regions of the input image that contribute most to the prediction of the target class. It computes a weighted sum of the feature maps, where the weights represent the importance of each feature channel to the prediction. To eliminate the negative contributions, a Rectified Linear Unit (ReLU) is then applied, highlighting more specific regions that positively support the class of interest (Selvaraju *et al.*, 2017).

Application in 3D: Natekar et al. (Natekar, Kori and Krishnamurthi, 2020) successfully applied Grad-CAM to a 3D U-Net for brain tumour segmentation, as it was originally designed for 2D images. Their research showed that 3D Grad-CAM can localise tumours based on the aggregated gradients across volumetric feature maps. However, it has a critical limitation which is the resolution trade-off: because Grad-CAM uses feature maps from deep layers, the resulting heatmaps are often coarse and may bleed into surrounding tissue, reducing the voxel-level precision of the segmentation mask itself (Natekar, Kori and Krishnamurthi, 2020).

Local Interpretable Model-Agnostic Explanations (LIME) takes a different approach. LIME treats the model as a function only, without using internal gradients.

Mechanism: LIME modifies the model's predictions and observes the change by masking out random superpixels (segments). It then fits a simple, interpretable model to these disrupted samples to predict the model's behaviour locally around a specific image (Ribeiro, Singh and Guestrin, 2016).

Evaluation in 3D: While LIME is highly recommended for 2D image segmentation, its performance in 3D is computationally expensive. According to Agrawal et al. (Agrawal *et al.*, 2025), LIME requires thousands of forward passes to generate stable explanations. For data like 3D images which contain millions of voxels, generating valid superpixels and running sufficient distributions to achieve significant statistics is often too slow for real-time clinical workflows. Additionally, defining the "superpixels" in a 3D brain volume is quite difficult, frequently leading to unstable explanations that vary between runs.

SHAP (SHapley Additive exPlanations) provides a theoretically sound substitute based on cooperative game theory. Each feature (voxel) is given an "importance value" that represents its marginal contributions to the prediction, averaged over all possible combinations of features (Ribeiro, Singh and Guestrin, 2016).

Strengths and Weaknesses: SHAP provides the most mathematically consistent explanations, by conforming the summation of each feature's attributes to be equal to the model's output. However, it has extreme computational costs in high-dimensional spaces. "DeepSHAP," an approximation method, has been used to accelerate this process, but research suggests that SHAP values can mislead the explanation in deep networks, as they may violate axioms when features are highly correlated, which is often common in statistically coherent MRI data (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

2.3.3 Comparative Synthesis and the Fidelity Gap

All three methods have their strengths and weaknesses, but Grad-CAM remains the best performing method for 3D medical image segmentation because of its computational efficiency (only one backward pass is needed) and its capacity to make use of the spatial information present in CNN feature maps (Natekar, Kori and Krishnamurthi, 2020).

However, a critical "fidelity gap" exists, as existing measures frequently show significant discrepancy between the predicted and ground truth segmentation (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025). This discrepancy is exacerbated by the standard practice of analysing 3D attention maps as static 2D slices, a method that hides volumetric coherence and increases cognitive load for clinicians. Therefore, closing the "trust gap" calls for a paradigm shift towards immersive virtual reality, which provides the high-dimensional, intuitive visualisation needed to properly understand these deep volumetric insights, rather than simply algorithmic improvements.

2.4 Immersive Visualisation in Medical Imaging

The interpretation of volumetric medical data has historically been analysed on 2D screens, which causes a lot of cognitive stress. According to Khedir et al. (Khedir *et al.*, 2025), traditional slice-by-slice navigation requires clinicians to mentally reconstruct volumetric architecture. This process requires time and is prone to spatial errors when assessing the depth of the tumours relative to prominent brain regions. Even though 3D Convolutional Neural Networks (CNNs) can process this volumetric data, the representation of their outputs remains largely stuck in 2D.

Emerging research suggests that immersive visualisation such as Virtual Reality (VR) and Augmented Reality (AR) can bridge this gap. VR provides **stereoscopic depth perception** and **6-Degrees-of-Freedom (6DoF)** interaction which goes well beyond standard monitors. This allows users to view the "saliency maps" not as flat overlays, which are generated by XAI, but as volumetric clouds suspended in 3D space. The NeuroXAI framework has shown that immersive environments significantly improve clinicians' ability to localise pathology and understand complex neural connectivity compared to standard desktop viewers (Zeineldin *et al.*, 2022).

The application of VR extends well beyond passive viewing to active surgical planning. Defining tumour boundaries in neuro-oncology is critical. Recent advancements in Augmented Reality (AR) enable surgeons to simulate trajectories and visualise "risk maps" created by deep learning models before entering the operation theatre (Zeineldin, 2023; Khedir *et al.*, 2025). For situations like complex glioma, where 2D scans may not represent the tumour with key blood arteries, this tool is a crucial resection corridor.

Additionally, the idea of interactive AI is essential to current medical systems. Static heatmaps (like standard Grad-CAM) offer **"take it or leave it"** explanations. VR environments, on the other hand, make **Human-in-the-Loop (HITL)** interaction easier. Immersive interfaces enable clinicians to interrogate the model, e.g., by digitally "erasing" a portion of the input volume to identify prediction changes (Orsmaa *et al.*, 2025). This interactivity transforms XAI from a static report into a dynamic conversation between the clinician and the AI, allowing the correction of model errors in real time (Orsmaa *et al.*, 2025).

To implement these immersive systems requires robust technological pipelines. Standard medical formats (DICOM/NIfTI) are not supported by native game engines like Unity. However, platforms like 3D Slicer have filled this bridge. The SlicerVR extension allows volume rendering of medical scans in VR headsets without any data conversion loss. Despite this, integrating XAI with VR still continues to be a major difficulty. Most pipelines concentrate on 2D slice-based visualisation, which fails to convey the volumetric feature importance and limits the clinical utility of XAI in more in-depth scenarios (Zeineldin *et al.*, 2022). There is a need for distinct workflows that can consume a 3D CNN's attention weights and render them as interactive volumetric objects, as proposed in frameworks like DeepIGN (Zeineldin, 2023).

2.5 Synthesis and Research Gap

2.5.1 The Disconnect Between Model Accuracy and Clinical Utility

After addressing and reviewing the three most significant chapters of this review, a distinct contrast has been established. Models like the 3D U-Net and its variants have achieved tremendous success in segmenting brain tumours (Çiçek *et al.*, 2016; Milletari, Navab and Ahmadi, 2016). On the other side, the opacity of the "black box" remains a critical barrier to clinical trust (Neri *et al.*, 2023). Although methods like Grad-CAM have successfully illuminated the internal logic of these networks (Natekar, Kori and Krishnamurthi, 2020), current implementations suffer from three major limitations that this project aims to address.

2.5.2 Identified Gaps in Current XAI Frameworks

Having global achievements in 3D XAI based on research such as NeuroXAI, there is still something which is called an **"interactivity gap"**; current systems use static visualisation rather than active user contribution (Zeineldin *et al.*, 2022). This passivity leads to a serious methodological error that confuses feature relevance with model confidence. As mentioned in the DeepIGN study (Zeineldin, 2023), a network may exhibit high focal attention on a specific region while displaying high epistemic uncertainty. This could give a confident result but incorrect false positives. As a result, existing 3D XAI methods do not provide the required mechanisms for **interactive correction or uncertainty quantification**, leaving a functional gap where the user cannot distinguish between a confident prediction and a statistical guess, or change the model's focus (Abyasa and Rahmania, 2025).

Furthermore, the utility of these volumetric insights is severely constrained by the "cognitive friction" of traditional visualisation mediums. Khedir et al. (Khedir *et al.*, 2025) mentioned that analysing 3D volumetric pathology via slice-by-slice 2D navigation increases significant burden to mentally reconstruct spatial relationships from disconnected images. The mismatch between the 3D nature of the data and the 2D nature of the display limits the interpretability of the feature maps, as crucial depth signals are lost in translation. Therefore, a clear research gap exists for a unique framework that not only generates uncertainty-aware explanations but also represents them in an immersive virtual environment, to clarify whether or not engagement significantly lowers cognitive load compared to conventional medical workflows.

2.6 Conclusion

This literature review establishes that while 3D deep learning architectures, particularly the 3D U-Net, have achieved superior success in volumetric tumour segmentation, their deployment is critically hindered by the "black box" paradox (Neri *et al.*, 2023). Clinical workflows still remain detached even if explainable methods like Grad-CAM provide mathematical solutions for transparency. The reliance on static 2D slice-by-slice visualisation fails to convey the complex spatial resolution of 3D pathologies and is hindered by a significant cognitive load that limits clinical trust (Natekar, Kori and Krishnamurthi, 2020).

Consequently, this research focusing on bridging the “trust gap” requires a paradigm shift from passive observation to an interactive environment. By developing a virtual reality environment that incorporates uncertainty quantification with real-time feedback, this project aims to “illuminate the black box” effectively. This innovative method will ensure that high-performance AI is not just accurate but also understandable and useful for medical practice.

Chapter 3: Methodology

3.1 Introduction

This chapter delineates the methodological framework for developing an Explainable AI (XAI) system capable of both accurate brain tumor segmentation and transparent clinical decision-making. The central objective is to address the "black-box" limitation common in medical imaging AI by transforming a complex deep learning model into an interpretable system. This system aims to achieve high segmentation accuracy while providing clinicians with actionable insights into the underlying reasoning behind each prediction.

To ensure transparency, the framework integrates six complementary XAI methodologies, each validated against ground-truth annotations to confirm their clinical relevance. At the core of this system is SegResNet, a 3D residual encoder-decoder architecture designed to process full volumetric MRI data rather than individual 2D slices. This volumetric approach enables the model to effectively capture the spatial hierarchies across three critical tumor sub-regions: the Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

The research design follows a hybrid Agile/CRISP-DM methodology, strategically combining the systematic rigor required for medical data mining with the iterative flexibility necessary for deep learning experimentation. This dual approach ensures methodological robustness appropriate for safety-critical healthcare applications while accommodating the evolutionary nature of neural network development, which transitioned from initial rapid prototyping on a 250-patient subset to full-scale training on the complete BraTS corpus.

The remainder of this chapter is structured as follows: Section 3.2 outlines the overall research approach and rationale; Section 3.3 details the technical implementation, including data preprocessing, model architecture, and XAI methods; Section 3.4 describes the research design and evaluation strategy; Section 3.5 discusses technical challenges and methodological limitations; and Section 3.6 addresses ethical considerations and data governance.

3.2 Approach

3.2.1 Overall Approach and Rationale

This study creates a three-stage pipeline using the BraTS 2023 dataset, which consists of about 1,250 multi-modal MRI patient cases: (1) training a SegResNet 3D CNN for volumetric tumour segmentation; (2) integrating multiple XAI techniques to illuminate model decision-making; and (3) exporting results into a VR-compatible format for immersive clinical evaluation. This volumetric methodology captures the true three-dimensional character of brain tumours, producing more anatomically accurate segmentations of the Whole Tumour, Tumour Core, and Enhancing Tumour areas, in contrast to 2D slice-by-slice methodologies that sacrifice inter-slice spatial context (Milletari, Navab and Ahmadi, 2016; Çiçek *et al.*, 2016).

SegResNet was selected as the primary architecture due to its residual connections, which provide robustness against vanishing gradients and enable effective training of deep networks (He *et al.*, 2016). The architecture offers an optimal balance between performance and computational complexity, making it suitable for medical imaging applications where training data may be limited and model interpretability is essential.

The rationale for implementing multiple XAI methods stems from their complementary strengths. Gradient-based methods (Grad-CAM, Guided Backpropagation, LRP) provide different perspectives on feature importance, while perturbation-based approaches (Occlusion Sensitivity) offer model-agnostic explanations. Monte Carlo Dropout adds uncertainty quantification, enabling identification of regions where the model lacks confidence, a critical capability for clinical decision support.

3.2.2 Agile CRISP-DM Methodology and Its Use

Deep learning model training is very exploratory and unpredictable, making traditional waterfall development lifecycles inappropriate. A more flexible approach is required due to the intricacy of neural network designs, as well as the requirement for substantial hyperparameter adjustment and iterative improvement. The Cross-Industry Standard Process for Data Mining's (CRISP-DM) structured phases are combined with Agile development's rapid iteration and evolutionary prototyping concepts in this study's hybrid Agile/CRISP-DM approach.

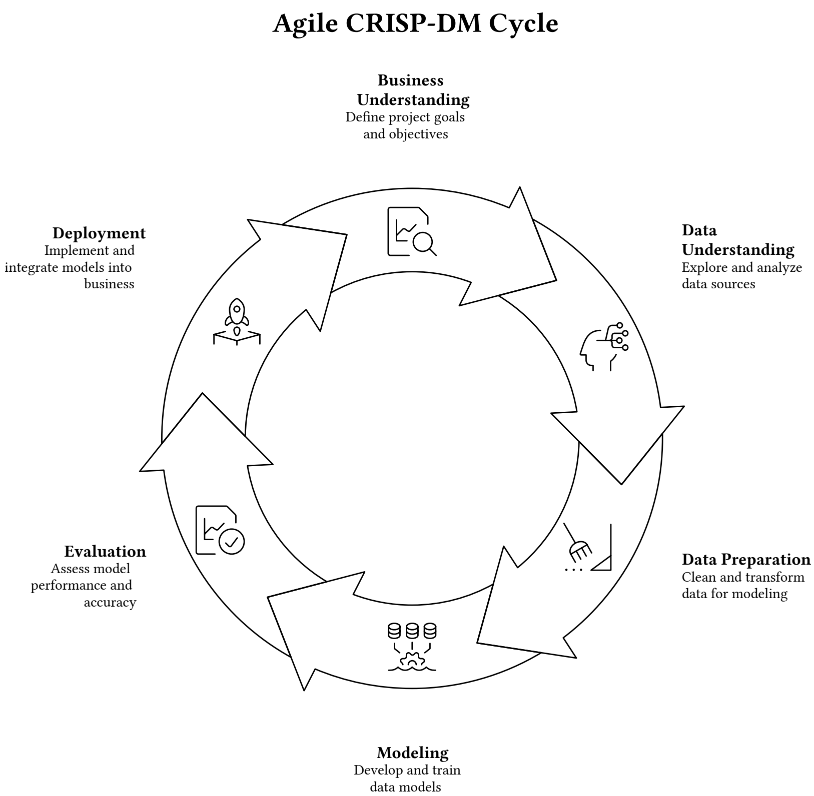


Figure 2: The Agile CRISP-DM methodology employed in this research, showing the iterative cycles between the six core phases.

Figure 2 illustrates the six interconnected phases of the methodology, with dashed arrows indicating the iterative nature of the development process. Each phase feeds into the next, but the Agile approach allows for rapid cycling back to earlier phases when issues are discovered or improvements are identified.

The framework, which was created using an iterative CRISP-DM process, was continuously validated against strict clinical criteria (Dice, HD95, IoU, sensitivity, and specificity) while containing Explainable AI components that were verified by saliency overlap and uncertainty quantification. Although the system is still an experimental research prototype awaiting clinical deployment, the pipeline produces standardised NIfTI outputs, containing segmentations, attention heatmaps, and variance arrays for direct 3D Slicer VR visualisation.

The Agile CRISP-DM approach ensured an organized, rapid-iteration development loop while maintaining the stringent robustness required for safety-critical medical machine learning components. Regular sprint reviews allowed for course correction, and the modular architecture enabled independent validation of each component.

3.3 Implementation

3.3.1 Data Acquisition and Preprocessing

The BraTS 2023 dataset, provided through the MICCAI Brain Tumor Segmentation Challenge, serves as the foundation for this research. The dataset comprises approximately 1,251 patient cases, each containing four co-registered MRI modalities: T1-weighted (T1), T1-weighted with Gadolinium contrast enhancement (T1c), T2-weighted (T2), and T2 Fluid-Attenuated Inversion Recovery (FLAIR). All volumes have been resampled to an isotropic resolution of 1 mm^3 and skull-stripped.

The tumor labels in BraTS follow a standard annotation scheme where each voxel is classified as background (0), necrotic tumor core (1), peritumoral edema (2), or enhancing tumor (3). For this research, these labels are converted into three binary channels representing clinically relevant regions: Whole Tumor (WT = necrosis + edema + enhancing), Tumor Core (TC = necrosis + enhancing), and Enhancing Tumor (ET = enhancing only). The preprocessing pipeline optimizes raw MRI volumes for efficient 3D model training through four core operations: foreground cropping reduces spatial dimensions from 240^3 to 160^3 voxels by removing extraneous background; channel-first formatting restructures data to [4, 160, 160, 160] for PyTorch compatibility; per-channel z-score normalization standardizes intensities across modalities using only brain tissue statistics; and random spatial augmentation (training only) applies geometric and intensity transforms such as rendom flips, adding noice etc. to enhance model robustness.

3.3.2 SegResNet Model Architecture

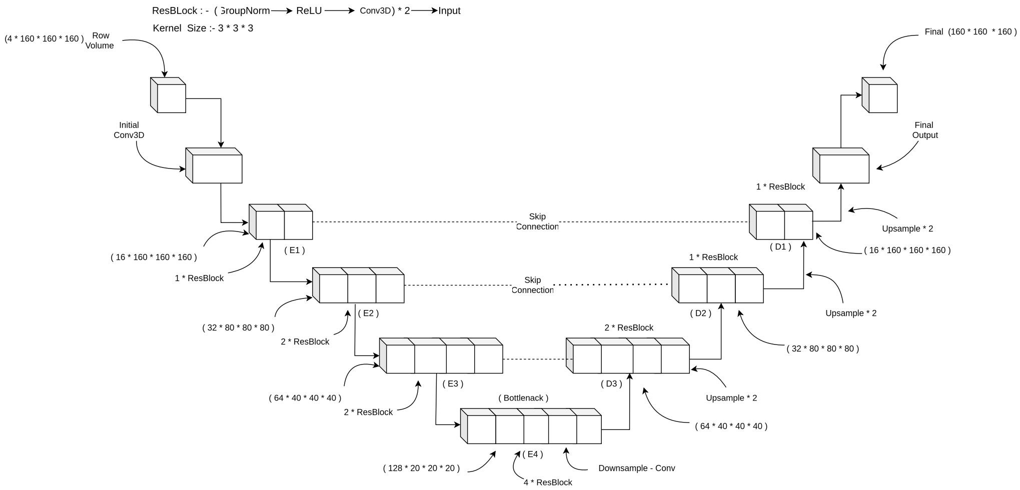


Figure 3: The SegResNet architecture, showing the encoder-decoder structure with residual connections and skip connections.

The implementation utilizes SegResNet, a 3D encoder-decoder architecture designed for volumetric segmentation that processes entire MRI volumes to capture global spatial dependencies. The encoder progressively downsamples input modalities through four levels, increasing filters from 32 to 256 channel bottleneck, using strided convolutions that reduce resolution from 160^3 to 20^3. Every layer is constructed from Residual Blocks (ResBlocks) featuring 3×3×3 kernels, GroupNorm, and ReLU activations. These blocks utilize skip connections to prevent vanishing gradients and ensure training stability by learning residual mappings rather than opaque transformations.

The bottleneck layer offers a compressed, globally-aware representation well-suited for Grad-CAM analysis. The decoder reverses the encoder via trilinear upsampling and skip connections, recovering fine-grained spatial detail for the final 160^3 outputs. To mitigate severe class imbalance particularly the Enhancing Tumor (ET) region at merely 5% volume, the model employs DiceFocalLoss ($\gamma = 2.0$) to emphasize misclassified voxels. A dropout rate of 0.1 serves dual purposes: regularization and enabling MC Dropout for uncertainty quantification in clinical review.

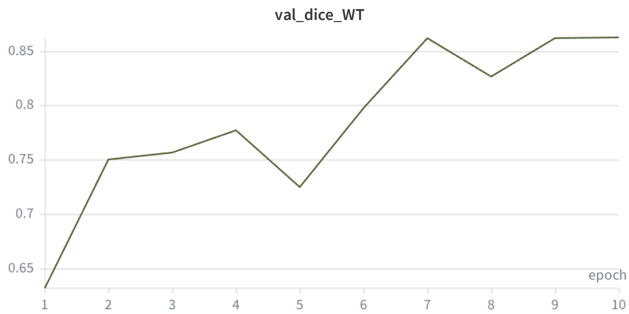
3.3.3 Evolutionary Prototyping Phase

Before committing to full-scale training, an evolutionary prototyping phase was conducted using a 250-patient subset randomly sampled from the BraTS training data. This phase served multiple purposes: validating the complex 3D data pipeline, rapidly testing candidate architectures, and verifying hardware memory limits.The initial prototype was implemented using an 2 NVIDIA GeForce RTX 2080 Ti (11 GB VRAM each), which allowed for training across 10 epochs in approximately 1 hours 2 min. The dataset was split into 200 training cases, 25 validation cases, and 25 test cases for this phase.

The prototype results demonstrated that SegResNet provided the optimal performance-to-complexity ratio among the tested architectures. The 250-patient subset achieved promising results on the prototype test set with Dice scores of 0.80 for the Whole Tumor (WT), 0.55 for the Tumor Core (TC), and 0.34 for the Enhancing Tumor (ET), confirming the viability of the approach for full-scale implementation. After validating this core functionality, the system was refactored into its final modular, configuration-driven PyTorch framework.

(a) Training Loss

(b) Batch Loss



(c) Validation Whole Tumor (WT) Dice

Figure 4: Training metrics for the 250-patient evolutionary prototype.

3.3.4 Implementation Pipeline

The implementation follows a modular architecture designed for reproducibility and extensibility, organized into several key components. YAML-based configuration files manage all hyperparameters and paths, supporting accessible parameter sweeps across the experiments. The core data pipeline (`src/data`) manages dataset scanning, patient splitting, and MONAI transform pipelines, while the model factory (`src/models`) dynamically instantiates model architectures, loss functions, and optimizers. Operating in tandem, the training framework (`src/training`) orchestrates the complete training lifecycle, integrating automatic mixed precision, automated checkpointing, and tracking via Weights & Biases. Finally, the explainability aspect is controlled by the XAI suite (`src/xai`), which standardizes the generation of saliency maps and uncertainty metrics, supported directly by the evaluation tools (`src/evaluation`) responsible for computing both segmentation and novel XAI-specific metrics and exporting them to CSV configurations.

The end-to-end workflow proceeds through seven stages: (1) configuration loading and data initialization, (2) model training with validation, (3) test set inference, (4) XAI map generation for all methods, (5) metric computation and CSV export, (6) NIfTI volume export for 3D Slicer, and (7) VR visualization preparation.

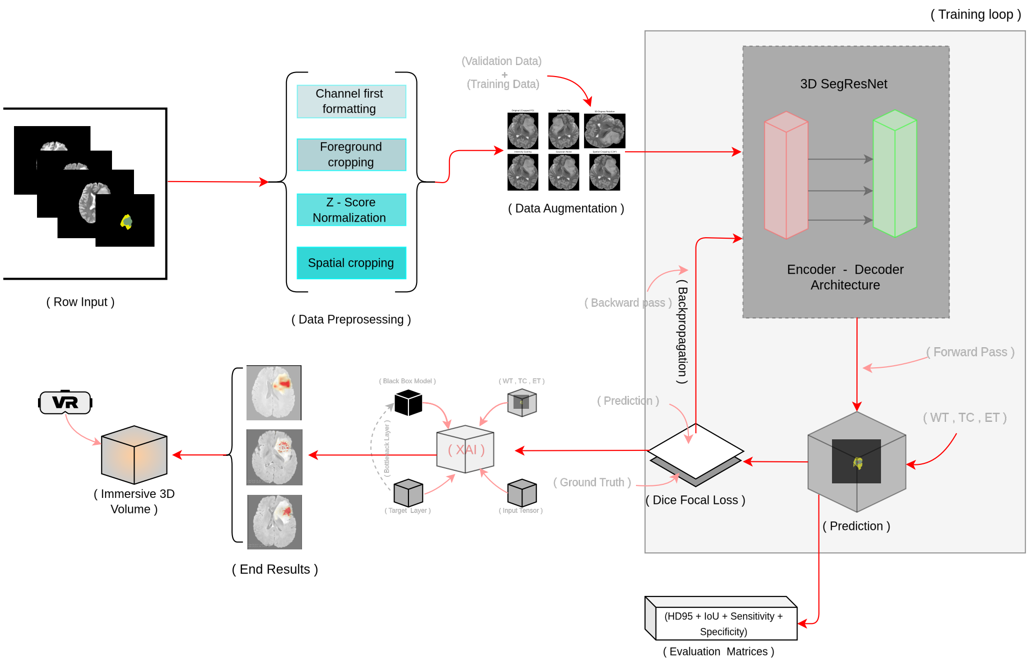


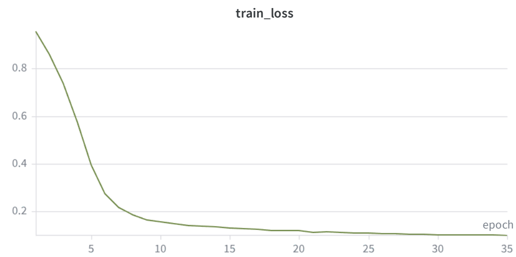
Figure 5: The implementation pipeline, showing the modular architecture and end-to-end workflow.

3.3.5 Full-Scale Training Setup

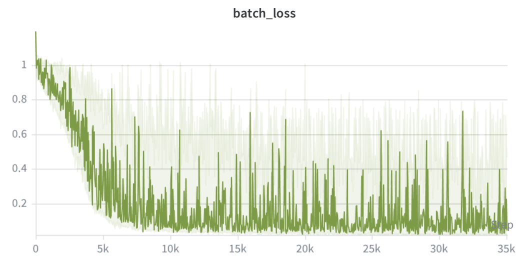
The final training configuration was optimized for a single high-memory GPU (RTX 5880 Ada, 48 GB VRAM) with Automatic Mixed Precision (AMP) enabled to maximize training efficiency. Key hyperparameters include:

Parameter	Value
Batch Size	1 (limited by 3D volume memory)
Epochs	35
Learning Rate	5×10^{-5}
LR Scheduler	Cosine Annealing
Optimizer	AdamW (weight decay 1×10^{-5})
Loss Function	DiceFocalLoss
Focal Gamma	2.0
Mixed Precision	Enabled

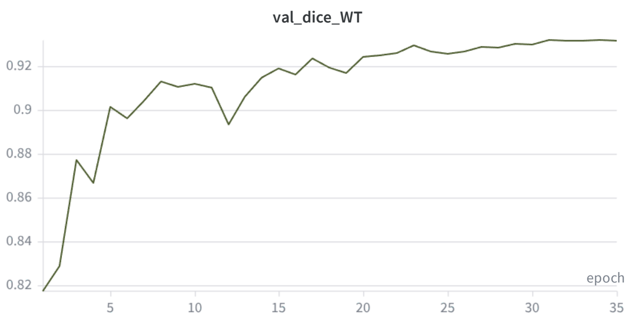
Table 1: SegResNet training hyperparameters for BraTS 2023 full dataset.



(a) Training Loss



(b) Batch Loss



(c) Validation Whole Tumor (WT) Dice

Figure 6: Training metrics for the full-scale SegResNet model (1,251 patients).

The dataset was split into 1,000 training cases, 125 validation cases, and approximately 126 test cases. The validation set was used for hyperparameter tuning and early stopping, while the test set was reserved for final evaluation and XAI analysis. All splits were performed deterministically using a fixed random seed (42) to ensure reproducibility.

Training progress was monitored using Weights & Biases (W&B) for experiment tracking, with metrics logged at every iteration. Best model checkpoints were selected based on validation Dice score, and the final model was evaluated on the held-out test set to report unbiased performance estimates.

3.3.6 XAI Module Implementation

The XAI framework implements six complementary explanation techniques, each providing unique insights into model decision-making:

Grad-CAM (Gradient-weighted Class Activation Mapping) computes per-feature-map importance weights by analyzing gradients flowing into the final convolutional bottleneck layer. This provides class-discriminative localization indicating regions critical for predicting a specific tumor sub-region. For a target class c and feature map k, the importance weight α_k^c is calculated by global-average-pooling the gradients across the spatial dimensions:

$$\alpha_k^c = \frac{1}{Z} \sum_x \sum_y \sum_z \frac{\partial y^c}{\partial A_{x,y,z}^k}$$

where y^c is the spatially-averaged output logit, $A_{x,y,z}^k$ represents the feature map activation, and Z is the total number of spatial elements. The gradient term $\frac{\partial y^c}{\partial A_{x,y,z}^k}$ mathematically represents the sensitivity or "importance" of the target class score y^c with respect to a specific pixel in the k-th feature map. By computing this partial derivative via backpropagation, the model quantifies exactly how much a change in that specific convolutional feature would objectively influence the final class probability. A linear combination of these feature maps is then computed, followed by a ReLU activation to preserve only features that positively support the target class:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right)$$

The resulting heatmap is upsampled to the original input resolution, providing class-specific but coarse spatial localization.

Guided Backpropagation (GBP) adds an additional guidance signal to the standard backpropagation algorithm to highlight areas where significant features are present at a highly detailed, voxel-level scale. When passing gradients backward through a ReLU unit at layer l, GBP suppresses the flow of negative gradients by enforcing two conditions: the original forward activation must have been positive, and the incoming backward gradient must also be positive. The guided gradient R_i^l propagated to neuron i is formulated as:

$$R_i^l = \left(f_i^l > 0\right) \cdot \left(R_i^{l+1} > 0\right) \cdot R_i^{l+1}$$

where f_i^l is the forward activation and R_i^{l+1} is the incoming gradient from the higher layer. Because both conditions act as binary indicator functions, purely positive signal paths are isolated. This generates a full-resolution (160^3) saliency map with sharp edges and fine detail, although it is not inherently class-discriminative.

Guided Grad-CAM fuses the complementary strengths of the previous two methods to achieve visualizations that are both high-resolution and class-specific. It masks the high-frequency pixel details produced by GBP using the coarse, class-discriminative localization of Grad-CAM through element-wise multiplication:

$$L^c_{\text{Guided Grad-CAM}} = L^c_{\text{Grad-CAM}} \cdot L_{\text{Guided Backpropagation}}$$

This process effectively masks out the fine-grained features that are irrelevant to the target class, resulting in high-resolution attributions that accurately highlight the structural features directly responsible for the model's prediction.

Layer-wise Relevance Propagation (LRP) operates on a deep Taylor decomposition "conservation principle," attributing a literal relevance score to each input voxel such that their sum equals the final model prediction score. Due to the architectural complexity of SegResNet, which encompasses numerous skip connections and specialized GroupNorm layers, developing manual backward hooks for every layer block is prohibitively complex. Therefore, the implementation utilizes the "Input × Gradient" approximation. For deep neural networks predominantly employing ReLU activations, this proxy is mathematically equivalent to the ε -LRP rule. The relevance R_i assigned to the i-th voxel of the input x_i concerning the spatial model prediction score $f(x)$ is formulated as:

$$R_i \approx x_i \cdot \frac{\partial f(x)}{\partial x_i}$$

This approach directly circumvents architectural bottlenecks, yielding an ultra-high resolution (160^3) relevance map that quantifies which specific structural features explicitly drove the network's prediction.

Occlusion Sensitivity serves as a purely empirical, model-agnostic perturbation technique. It bypasses internal gradient calculations altogether, instead physically testing the 3D CNN's structural reliance by obscuring data. The implementation employs a 3D sliding window defined by a $16 \times 16 \times 16$ spatial occluding patch that moves across the target volume with a stride of 8 voxels. At each step, the occluded region within the input MRI modalities x is replaced with a baseline value of zero, creating an occluded volume x_{occluded}. For a target class c, the spatial sensitivity score S_c evaluated at the occluding window's center coordinate (i, j, k) measures the absolute drop in the model's confidence f_c:

$$S_c(i, j, k) = f_c(x) - f_c(x_{\text{occluded}})$$

Because the sliding window utilizes a stride of 8 across a 160^3 input space, the resulting sensitivity map is initially evaluated at a dimension of 20^3. It is subsequently reconstructed and upsampled back to 160^3 via trilinear interpolation. This systematic validation step quantifies the clinical necessity of explicit brain regions on an undeniable, empirical level.

Monte Carlo (MC) Dropout quantifies confidence rather than providing structural attribution. Because traditional inference is deterministic, this implementation introduces test-time stochasticity by forcing SegResNet's built-in dropout layers (at a probability of $p = 0.1$) to remain active during the evaluation phase. Generating $N = 20$ stochastic forward passes for an identical volumetric input produces a probability distribution over the predicted segmentations. Given the predicted output probability p_n from the n-th forward pass, the stabilized mean prediction $\bar{p}$ and its corresponding voxel-wise predictive variance (uncertainty map) σ^2 are evaluated as:

$$\bar{p} = \frac{1}{N} \sum_{n=1}^{N} p_n$$

$$\sigma^2(x, y, z) = \frac{1}{N} \sum_{n=1}^{N} \left(p_{n(x,y,z)} - \bar{p}(x, y, z)\right)^2$$

Where high variance occurs, it highlights "brittle" transition zones or anatomically ambiguous boundaries where the network remains unconfident. Comparing LRP attributions against these uncertainty maps isolates clinically critical structural dependencies that carry significant diagnostic risks.

All XAI outputs are saved as NIfTI volumes organized per patient and per method, ready for loading into 3D Slicer or VR visualization tools. The implementation ensures spatial alignment between saliency maps and ground truth labels by computing metrics inline during generation, using the same preprocessed coordinate space.

3.4 Research Approach

3.4.1 Research Design

This study employs applied, quantitative research using secondary medical imaging data. The experimental design evaluates one primary architecture (SegResNet) combined with six XAI techniques, measuring both segmentation performance and explanation validity against ground truth annotations.

The experimental unit is a single BraTS patient case. Independent variables include the choice of XAI method and the tumor sub-region being evaluated (WT, TC, ET). Dependent variables encompass segmentation metrics (Dice, HD95, IoU, Sensitivity, Specificity), XAI metrics (Pointing Game, Coverage, IoU, Weighted Dice), and uncertainty statistics (UAR, boundary ratios).

3.4.2 Evaluation Strategy

Model performance and internal reliability are evaluated systematically. To ensure clinical interpretability, the quantitative assessment is divided into mathematical evaluations for structural segmentation accuracy and subsequent evaluations for XAI attribution and uncertainty.

1. Segmentation Performance Metrics The core segmentation performance of SegResNet is benchmarked against the clinically-annotated test set using the following spatial metrics:

Metric Name	Formula	Description / Function
Dice Score (DSC)	$\frac{2\lvert A \cap B\rvert}{\lvert A\rvert + \lvert B\rvert}$	Measures volumetric overlap between prediction (A) and ground truth (B) from 0 to 1. Primary BraTS benchmark.
HD 95th Percentile	$P_{95}(d(a, B) \cup d(b, A))$	Measures 95th percentile surface boundary error in millimeters. Crucial for assessing margin precision.
Intersection over Union	$\lvert A \cap B\rvert / \lvert A \cup B\rvert$	A stricter spatial overlap metric (Jaccard index) penalizing localized false positives more heavily than Dice.
Sensitivity (Recall)	$\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$	The true positive rate. Measures the fraction of actual tumor voxels correctly identified to prevent missed diagnoses.
Specificity	$\frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}$	The true negative rate. Measures reliability in excluding healthy tissue, directly preventing false alarms.

Table 2: Summary of quantitative metrics used for evaluating 3D volumetric segmentation accuracy.

2. Explainability and Uncertainty Metrics Beyond structural accuracy, the generated gradient-based visual attributions and the MC Dropout variance maps (σ^2) are quantitatively assessed utilizing custom validation metrics:

Metric Name	Formula	Description / Function
Pointing Game (PG)	$\begin{cases} 1 & \text{if} \arg\max(L) \in G \\ 0 & \text{otherwise} \end{cases}$	Binary test returning 1 if the highest activation peak in the saliency map L falls inside the tumor region G.
Saliency Coverage	$\frac{\sum_{x \in G} L(x)}{\sum_{x} L(x)}$	Calculates the percentage of total salient relevance mass concentrated inside the true tumor boundary.
Saliency IoU	$\lvert L_{\text{thresh}} \cap G \rvert / \lvert L_{\text{thresh}} \cup G \rvert$	Evaluates shape overlap between a thresholded Boolean heatmap (L_{thresh}) and the ground truth mask (G).
Weighted Dice	$\frac{2 \sum L(x) \cdot G(x)}{\sum L(x) + \sum G(x)}$	Treats continuous heatmaps as soft predictive weights, modeling organic relevance without binary thresholds.
Uncertainty Area Ratio	$\frac{\sum_{x \in G} \sigma^2(x)}{\sum_{x} \sigma^2(x)}$	Computes the ratio of predictive uncertainty (variance σ^2) trapped inside the pathogenic ROI versus total volume.
Saliency-Uncertainty	$\frac{\text{Cov}(L, \sigma^2)}{\sigma_L \sigma_{\sigma^2}}$	Pearson correlation isolating "brittle" behaviors where high relevance (L) intersects with high variance (σ^2).

Table 3: Summary of metrics utilized to evaluate XAI map alignment and model uncertainty.

3.4.3 Link to Research Questions

This methodological design directly addresses the research questions posed in Chapter 1:

- **Segmentation Accuracy** is measured by traditional metrics (Dice, HD95), answering whether the model achieves clinically acceptable performance.
- **Model Attention Alignment** is assessed by XAI metrics (Pointing Game, Coverage), answering whether the model "looks at the right place for the right reasons."
- **Reliability Characteristics** are quantified by uncertainty metrics, answering where the model is confident versus uncertain.

By combining these evaluation dimensions, this research can argue not only that the model is accurate, but also that its decision-making process is clinically interpretable and trustworthy.

3.5 Challenges and Limitations

This research navigated several primary technical and methodological constraints:

Computational and Architectural Constraints: The memory-intensive nature of 3D CNNs mandated a limited batch size (1) and a 160^3 volumetric ROI, affecting training efficiency (approx. 24 hours for 35 epochs). XAI generation proved exceptionally demanding; Occlusion

Sensitivity required approximately 1,000 forward passes per patient. Architecturally, integrating gradient-based XAI methods required strict hook isolation. For instance, simultaneous Grad-CAM and Guided Backpropagation hooks inherently corrupt each other's gradients, demanding sequential, per-method computation workflows.

Generalizability and Domain Shift: Methodologically, relying exclusively on an isolated dataset (BraTS 2023) without external cross-institutional validation restricts the clinical generalization of the model. Its segmentation performance may degrade due to domain shift if exposed to MRI scans from alternate hospital scanners, protocols, or distinct neuropathologies.

XAI Construct Validity: Finally, while metrics like the Pointing Game and Weighted Dice effectively operationalize interpretability mathematically, they serve merely as academic proxies for clinical utility. Furthermore, the Virtual Reality (VR) application acts primarily as an exploratory visualization target; comprehensive studies involving practicing radiologists remain critical to validating the actual diagnostic efficacy of these immersive explanation maps.

3.6 Ethical Considerations

3.6.1 Data Privacy and Governance

This research uses only de-identified BraTS data that has been stripped of all personally identifiable information. Data handling practices include local secure storage with restricted access, no attempt at re-identification, and adherence to dataset usage terms. The BraTS challenge operates under appropriate ethical approvals for secondary use of medical imaging data in research.

3.6.2 Clinical Safety and Responsible Use

It is essential to emphasize that this system is a research prototype and has not been approved for clinical decision-making. The intended use is as a decision-support and research tool to explore model behavior, not as an autonomous diagnostic system. Risks include over-reliance on AI outputs and potential misinterpretation of saliency maps or uncertainty maps by non-expert users.

Clear documentation accompanies all outputs explaining what XAI maps represent and what they do not. The system is designed to augment, not replace, clinical expertise.

3.6.3 Bias and Generalisation

The BraTS dataset, while large and diverse, may contain biases related to geography, scanner types, and pathology distribution. Limited external validation means performance may change on under-represented populations. Future work should include broader, more diverse datasets to ensure equitable performance across demographic groups.

3.6.4 Ethical Use of VR for Medical Imaging

Immersive visualization may amplify visual cues (e.g., bright saliency hotspots) and affect clinician perception. Safeguards for future deployment include clear documentation on XAI map interpretation, training materials for clinicians, and emphasis on combining AI insights with expert judgment rather than relying solely on automated outputs.

Chapter 4: Results

This chapter presents a comprehensive appraisal of the 3D Convolutional Neural Network (SegResNet) and its efficacy. The results are organized into three sequential analyses: an evaluation of the model's predictive segmentation performance, the interpretability of its decisions via Explainable Artificial Intelligence (XAI) techniques, and the subsequent visualization of these complex metrics within an immersive Virtual Reality (VR) framework.

4.1 Prediction and Segmentation Performance

The foundational technical phase of this study necessitated the development of a neural network architecture capable of accurately localizing brain tumor subregions, serving as a prerequisite for subsequent explainability analyses. Given the substantial histopathological heterogeneity and complex morphological variations characteristic of neoplastic lesions (actual tumor), establishing a robust segmentation backbone was imperative to ensure the validity of downstream interpretability methods.

4.1.1 Progression and Metric Analysis (Baseline vs. Final)

This section presents a comparative analysis of model iterations developed through the Agile-based methodology previously described. Beginning with a baseline prototype (Version 1) trained on minimal data and culminating in a fully-specified model trained on 1,251 patients from the BraTS cohort, the evaluation metrics reveal substantial performance divergences between these developmental phases.

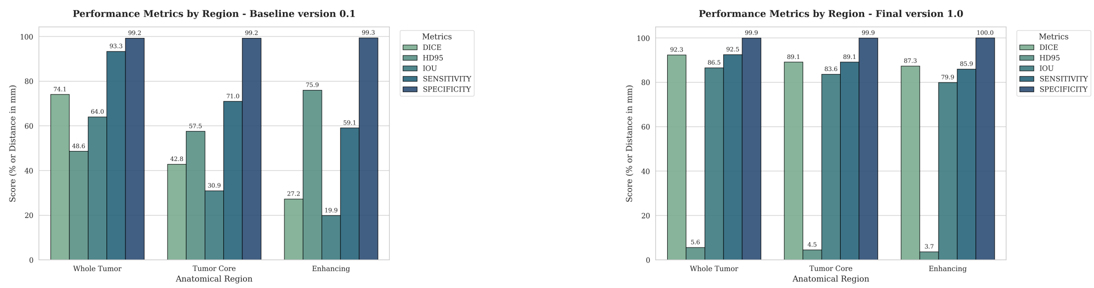


Figure 7: Performance comparison between the Baseline Prototype (left) and the Final Model trained on 1,251 patients (right). Note the massive improvement in the ET metric.

Data Scaling Comparison To understand how dataset volume directly influences performance, we compared the Dice scores across three versions: the initial baseline, an intermediate training run of 250 patients, and the final run utilizing the complete 1,251-patient dataset.

	Dice Score			HD95 (mm)		
Region	**Baseline**	**250 Pts**	**Final**	**Baseline**	**250 Pts**	**Final**
WT	0.741	0.908	0.923	48.63	7.23	5.60
TC	0.428	0.837	0.891	57.54	15.34	4.54
ET	0.272	0.742	0.873	75.91	15.25	3.66

Table 4: Test set Dice and HD95 progression across data scaling stages. Dice quantifies volumetric overlap (higher is better) while HD95 measures boundary error in millimetres (lower is better).

Table @tab:dice-hd95 presents segmentation metrics across training data scales (45 to 1,251 patients), utilizing Dice coefficients for volumetric overlap and 95th Percentile Hausdorff Distance (HD95) for boundary delineation. Whole Tumor segmentation exhibited asymptotic performance (Dice 0.908 to 0.923), suggesting efficient learning of macroscopic structures from limited data. In contrast, the Enhancing Tumor characterized by fragmented, heterogeneous morphology demonstrated substantial dependency on training volume, improving from 0.272 to 0.873. Corresponding boundary refinement was equally pronounced: ET HD95 decreased from 75.91 mm to 3.66 mm, progressing from clinically unusable deviations to sub-voxel precision requisite for safe neurosurgical planning.

Computational vs. Performance Trade-off

The escalation from 250 patients to 1,251 patients carried substantial hardware and temporal costs. Training iterations lasted exponentially longer on the GPU infrastructure. However, as the table indicates, the added computation was unequivocally justified. While general regions (WT) hit a point of diminishing returns in both Dice and HD95, the deeper feature extraction required to confidently segment the complex non-linear Enhancing Tumors and to tighten their boundary precision required the full breadth of the training corpus.

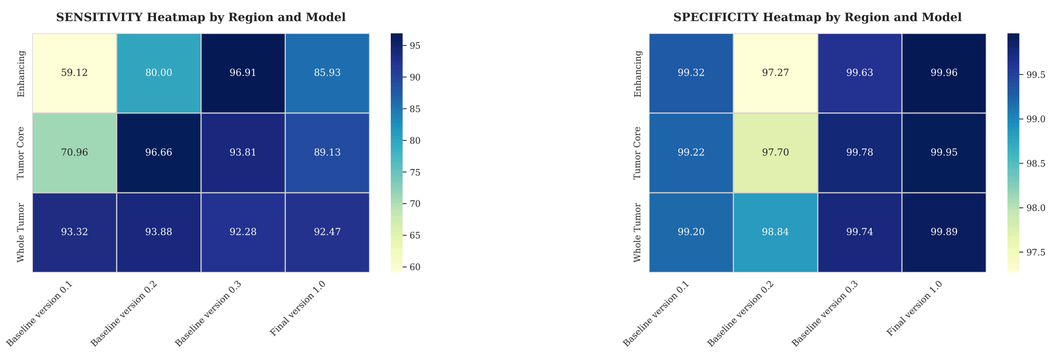


Figure 8: Sensitivity (left) and Specificity (right) heatmaps across model versions. Sensitivity measures the proportion of true tumor voxels correctly identified, while Specificity reflects the model's ability to correctly exclude healthy tissue.

4.1.2 Qualitative Visual Evaluation (2D & 3D Comparisons)

Morphological fidelity is still necessary for therapeutic value, even though quantitative metrics demonstrate computational validity. Precise spatial localisation of complex tumour borders is necessary for accurate subregion delineation. In order to confirm spatial concordance and anatomical accuracy, SegResNet predictions are compared to expert-annotated ground truth using multi-planar (2D) and volumetric (3D) visual evaluations.

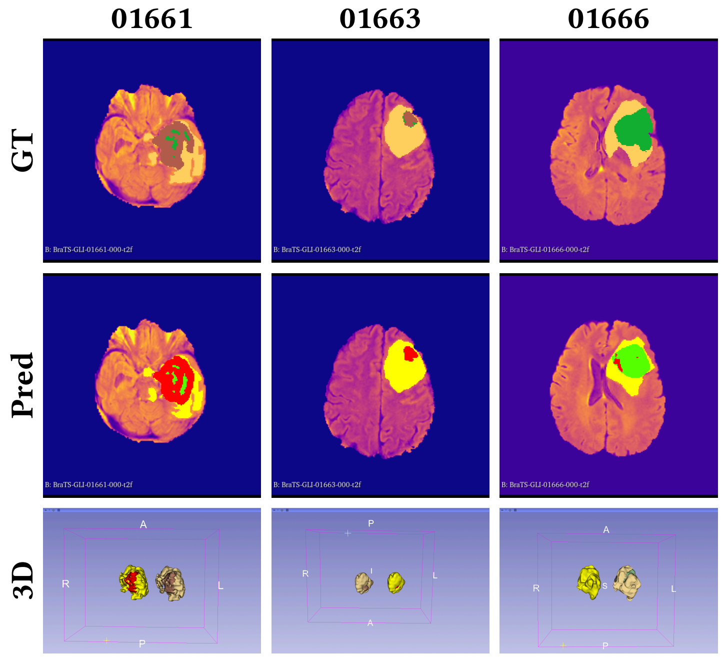


Figure 9: Qualitative comparison of patients 01661, 01663, and 01666. Row 1: Ground Truth (GT). Row 2: Model Prediction (Pred). Row 3: 3D volumetric rendering of the predicted segmentation.

2D Alignment Analysis When examining the 2D planes, the model's predicted boundary layers flawlessly align with the True Ground Truth masks. As established, the 3.66 mm precision in HD95 manifests visually here: the predicted layers perfectly hug the edges of the actual tumor on the base MRI scans rather than loosely bleeding into healthy tissue.

3D Morphological Analysis While conventional slice-based analysis obscures the true volumetric extent of neoplastic growth, three-dimensional reconstruction (Panel D) reveals the complete spatial topography and structural depth of the lesion. SegResNet's native volumetric processing capability generates high-fidelity 3D masks that facilitate direct translation into immersive rendering environments. This dimensional accuracy establishes the essential foundation for the immersive visualization framework presented in Pillar 3.

4.2 Explainable AI (XAI) Interpretation

Having established clinically viable segmentation accuracy, the analysis shifts from how well the model performs to why it predicts specific sub-regions, addressing the "black box" opacity that undermines trust in neuro-oncological workflows (Neri *et al.*, 2023). Six complementary post-hoc techniques, taxonomised across three XAI paradigms (Bhati, Neha and Amiruzzaman, 2024), are deployed to interrogate this decision-making. Gradient-based attribution, comprising Grad-CAM for coarse class-discriminative localisation (Selvaraju *et al.*, 2017; Natekar, Kori and Krishnamurthi, 2020), Guided Backpropagation for voxel-level structural dependencies, and their Guided Grad-CAM fusion, generates multi-scale saliency. Decomposition-based relevance via Layer-wise Relevance Propagation redistributes output scores to individual voxels under a conservation principle. Perturbation-based validation through Occlusion Sensitivity provides model-agnostic empirical proof of structural reliance, while stochastic uncertainty quantification via Monte Carlo Dropout maps epistemic ambiguity at tumour boundaries. Together, these converging paradigms close the fidelity gap, ensuring predictions are not only accurate but transparent, verifiable, and reliability-aware.

Category	**Method**	**Resolution**	**Class-Specific**
Gradient-Based	3D Grad-CAM	Coarse (20^3)	Yes
Gradient-Based	Guided Backpropagation (GBP)	Full (voxel)	No
Gradient-Based	Guided Grad-CAM	Full (voxel)	Yes
Relevance-Based	LRP (Input × Gradient)	Full (voxel)	Yes
Perturbation-Based	Occlusion Sensitivity	Stride-upsampled	Yes
Uncertainty-Based	MC Dropout (20 passes)	Full (voxel)	Yes

Table 5: Summary of the six XAI techniques applied to the SegResNet model, categorised by mechanism, spatial resolution, and class discrimination capability.

4.2.1 XAI Evaluation Metrics: Addressing the Methodological Gap

Quantitative validation of explainability in 3D medical segmentation remains methodologically underdeveloped. Established literature predominantly evaluates saliency maps through **Pointing Game** accuracy and **Saliency Coverage**, binary or ratio-based measures that assess whether peak attention falls within the ground truth region or what fraction of total saliency mass concentrates inside the tumour boundary (Natekar, Kori and Krishnamurthi, 2020; Bhati, Neha and Amiruzzaman, 2024). While these metrics operationalise localisation fidelity, they suffer from critical limitations: Pointing Game reduces volumetric interpretability to a single voxel hit-or-miss test, and Coverage remains agnostic to spatial distribution, permitting diffuse, anatomically imprecise attention to score favourably.

Furthermore, **Saliency IoU**, which thresholds continuous heatmaps into binary masks before computing Jaccard overlap, introduces hard-thresholding artefacts that disproportionately penalise coarse-resolution methods such as Grad-CAM while discarding gradient intensity information essential for clinical nuance (Miró-Nicolau, Jaume-i-Capó and Moyà-Alcover, 2025).

To address this methodological gap, this research introduces **Weighted Dice**, a novel soft-metric adaptation that treats continuous saliency values as probabilistic membership weights rather than forcing premature binarisation. Weighted Dice is formulated as:

$$\mathrm{WD} = \frac{2 \cdot \sum (S \cdot G)}{\sum S + \sum G}$$

where $S \in [0, 1]$ denotes the continuous saliency distribution and G the binary ground truth. By treating the continuous saliency values directly as soft membership scores, Weighted Dice preserves full gradient intensity information, penalises both spatial misalignment and saliency leakage into healthy tissue, and remains stable across varying spatial resolutions. This constitutes a substantive contribution to 3D medical XAI methodology, enabling equitable comparison of coarse bottleneck attributions against full-resolution gradient maps without threshold-induced volatility.

4.2.2 The Bottleneck Resolution Problem: 3D Grad-CAM

Grad-CAM, conceived for 2D classification (Selvaraju *et al.*, 2017), was adapted to 3D segmentation by spatially averaging per-class logits for scalar backpropagation (Natekar, Kori and Krishnamurthi, 2020). This yields a coarse 20^3 bottleneck heatmap subsequently upsampled to native 160^3 introducing fundamental spatial fidelity loss in volumetric contexts. The critical question does trilinear upsampling inflate or deflate evaluation scores?

Identical activations were evaluated at both resolutions using Weighted Dice and Saliency IoU. As shown in Figure 10, Weighted Dice remains stable (± 0.02–0.04), confirming no systematic bias. Conversely, Saliency IoU exhibits volatile thresholding artefacts, validating Weighted Dice as the reliable metric for coarse-resolution attribution. Critically, Grad-CAM achieves 0% Pointing Game for Enhancing Tumour even at native resolution,a fundamental bottleneck limitation, not model incapacity.

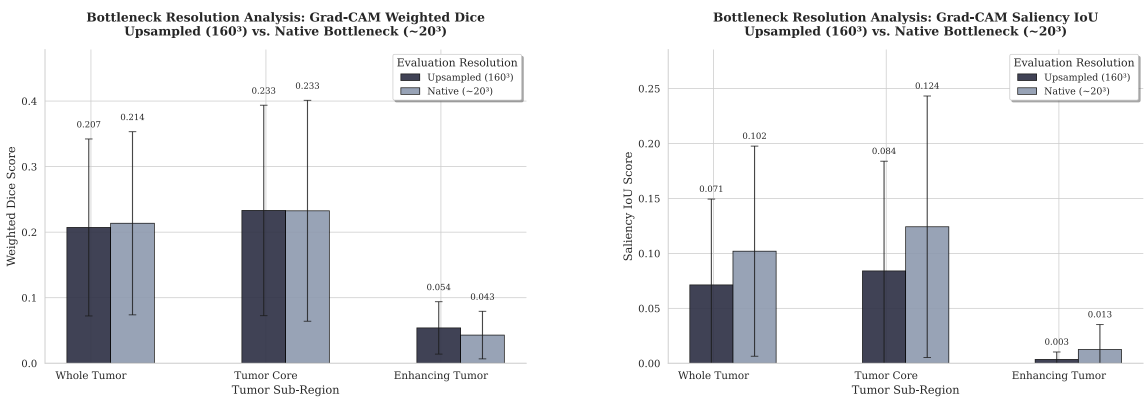


Figure 10: Bottleneck resolution analysis comparing Weighted Dice (left) and Saliency IoU (right) for Grad-CAM evaluated at upsampled 160^3 versus native 20^3 resolution. Weighted Dice remains stable across both resolutions, while Saliency IoU exhibits volatile swings due to hard thresholding artefacts.

4.2.3 Full-Resolution Gradient Attribution: Guided Backpropagation

Guided Backpropagation (GBP) modifies standard backpropagation by gating negative gradients at each ReLU, isolating purely positive signal paths to produce full-resolution (160^3) saliency maps. Unlike Grad-CAM's bottleneck-dependent coarse localisation, GBP operates at native voxel scale without class-specific weighting, revealing fine-grained structural dependencies directly from input-level features. As shown in Figure 11, GBP achieves 100% Pointing Game across all patients and tumour regions including Enhancing Tumour, where Grad-CAM fails entirely. While Saliency Coverage remains modest (0.12–0.44) due to distributed edge-highlighting rather than concentrated blob detection, this diffuse pattern serves as a critical sanity check: the model encodes tumor relevant features at the input pixel level, not merely in deep bottleneck abstractions, rendering every prediction traceable to real anatomical structure.

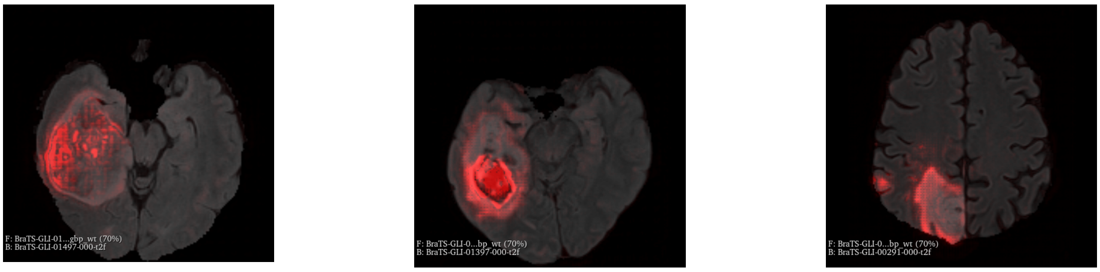


Figure 11: GBP saliency maps for patients 01497 (left), 01397 (centre), and 00291 (right). GBP achieves 100% Pointing Game across all patients and all tumour regions, including patient 00291, where Grad-CAM and Guided Grad-CAM produce zero activation.

4.2.4 Gradient Fusion: Guided Grad-CAM

Guided Grad-CAM fuses GBP's fine grained spatial detail with Grad-CAM's class-specific weighting via element wise multiplication of the full resolution GBP saliency and the upsampled Grad-CAM heatmap. For patient 01497, this recovers spatial information lost in Grad-CAM's bottleneck while preserving class-discriminative focus, tightly concentrating saliency within tumour boundaries.

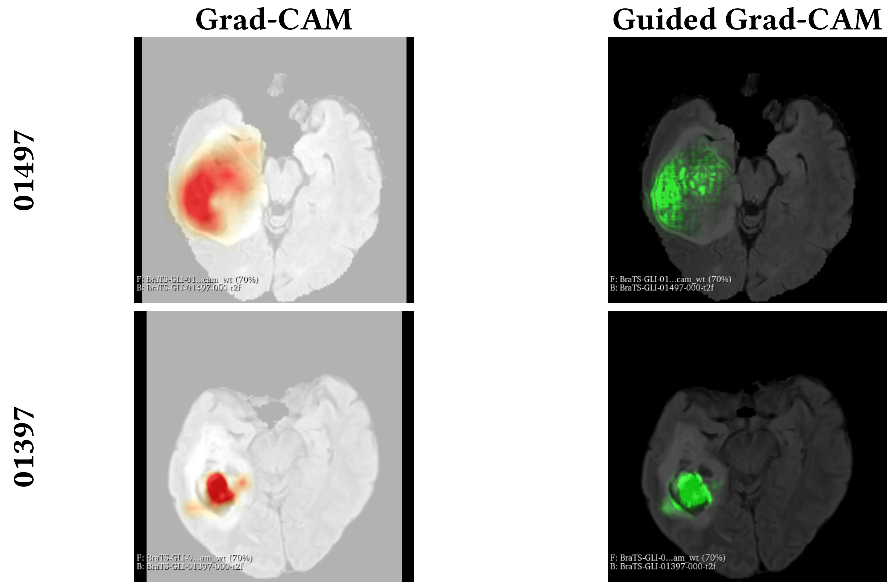


Figure 12: Grad-CAM (left column) versus Guided Grad-CAM (right column) for patients 01497 (top) and 01397 (bottom). The fusion recovers spatial detail lost in the bottleneck while preserving class-discriminative weighting, concentrating saliency tightly within tumour boundaries across varying morphologies.

Quantitatively, the fusion achieves the highest Saliency Coverage of all methods evaluated: 0.81–0.96 for Whole Tumour and 0.81–0.92 for Tumour Core, with nearly all saliency mass localized inside the tumour. Enhancing Tumour Coverage also rises substantially from Grad-CAM's near-zero to 0.03–0.38, confirming that the full-resolution GBP component restores detail Grad-CAM alone cannot capture.

However, the fusion inherits Grad-CAM's failure modes. For patient 00291, where Grad-CAM produces a zero activation map, the element-wise multiplication zeros out GBP's otherwise perfect signal, producing blank maps across all metrics. This critical limitation demonstrates why no single XAI method is sufficient for clinical deployment; multi-method evaluation is essential.

4.2.5 Relevance-Based Attribution: LRP (Input × Gradient)

Layer-wise Relevance Propagation (LRP), implemented here as Input × Gradient, backpropagates the model's output score to individual input voxels. Unlike Grad-CAM, which shows *where* the model attends, LRP reveals *what evidence* supports its predictions.

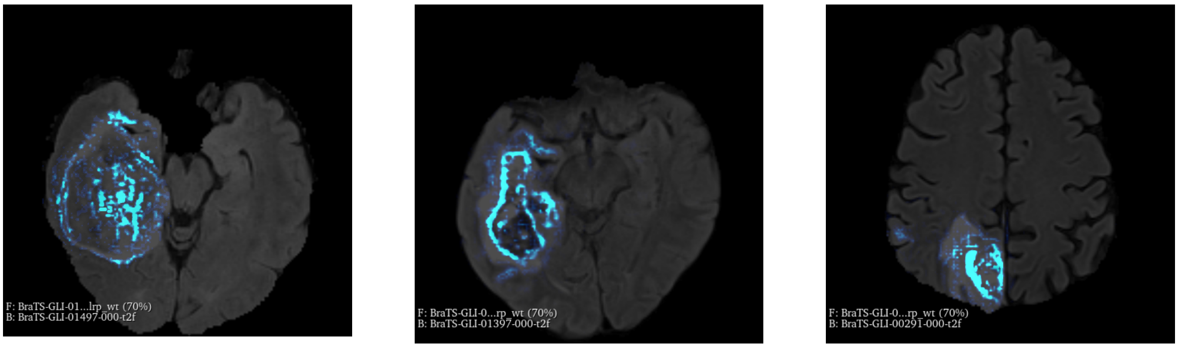


Figure 13: LRP saliency maps for patients 01497 (left), 01397 (centre), and 00291 (right). LRP achieves 100% Pointing Game across all patients and all regions, producing diffuse but correctly localised relevance distributions.

LRP achieves **100% Pointing Game across all five patients and three tumour regions**, including Enhancing Tumor, matching GBP's perfect localisation. Saliency Coverage of 0.34–0.84 indicates relevance extends beyond tumour boundaries into surrounding tissue context (e.g., peritumoural oedema for Whole Tumour predictions).

Low Saliency IoU (< 0.01) and moderate Weighted Dice (0.01–0.17) reflect LRP's diffuse nature rather than poor alignment. The consistent perfect Pointing Game confirms that predictive features remain co-located with tumour tissue across all spatial scales.

4.2.6 Perturbation-Based Attribution: Occlusion Sensitivity

Unlike gradient-based approaches, Occlusion Sensitivity requires no assumptions about model internals. It slides a 16^3 black patch across the input, recording confidence drops at each position to map model dependency empirically. This perturbation-based approach serves as the **gold-standard XAI validation**.

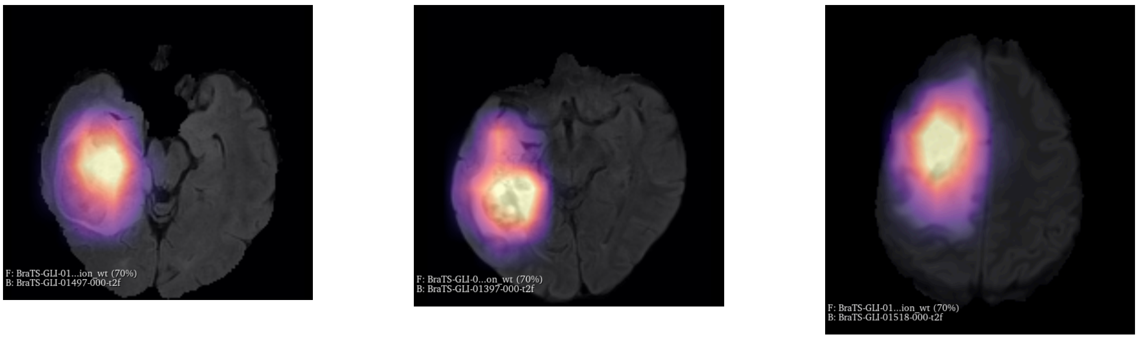


Figure 14: Occlusion Sensitivity heatmaps for patients 01497 (left), 01397 (centre), and 01518 (right). Occlusion achieves the highest Weighted Dice of all six methods, with patient 01397 ET achieving W.Dice = 0.35, the only method where ET localisation truly succeeds above 0.30

Occlusion achieves 100% Pointing Game and MSR Accuracy for Whole Tumour across all patients, and produces the highest Weighted Dice scores of all six methods: 0.38–0.46 for WT, 0.24–0.47 for TC, and 0.23–0.35 for ET. Most remarkably, patient 01397 achieves a Weighted Dice of 0.35 with PG = 1.0 for Enhancing Tumor, the only method where ET localisation succeeds above the 0.30 threshold. This result constitutes the single strongest piece of evidence that the SegResNet model genuinely relies on tumour voxels for its segmentation predictions, ruling out shortcut learning, texture bias, or dataset artefacts. The clinical trustworthiness of the segmentation is thereby validated through a mechanism entirely independent of gradient computation.

4.2.7 Uncertainty Quantification: MC Dropout

MC Dropout estimates model uncertainty through 20 stochastic forward passes, producing per-voxel variance maps that complement saliency methods. Six metrics quantify this: Uncertainty Area Ratio (UAR), Boundary Uncertainty Ratio, mean uncertainty inside/outside the tumour, LRP Weighted Dice, and Saliency-Uncertainty Correlation (Pearson r between LRP saliency and MC Dropout variance within the tumour mask). LRP was selected for correlation analysis because its native 160^3 resolution matches MC Dropout's output, avoiding interpolation artefacts inherent to Grad-CAM's 20^3 maps.

Patient 01497 – Low Uncertainty, Clear Boundaries

Region	UAR	Boundary Ratio	Unc Inside	Unc Outside	LRP Corr
WT	0.196	0.876	0.00131	0.00010	+0.020
TC	0.092	**0.967**	0.00122	0.00016	+0.000
ET	0.219	0.579	**0.01400**	0.00005	−0.065

Table 6: MC Dropout uncertainty metrics for patient 01497. Low UAR indicates confident segmentation. TC Boundary Ratio of 0.967 demonstrates that nearly all uncertainty concentrates at Tumour Core edges.

This case shows confident segmentation, with UAR ranging from 0.083 (TC) to 0.226 (WT). Boundary ratios of 0.906 (WT) and 0.964 (TC) indicate uncertainty concentrates at tumour edges, while the ET boundary ratio of 0.503 reflects more distributed uncertainty. ET displays the highest internal variance (0.0130) relative to background (0.00007), correctly identifying enhancing tissue as the most challenging sub-region. Saliency-Uncertainty correlations are negligible (−0.055 to +0.021), suggesting relevance and uncertainty are largely decoupled.

Patient 01397 – High Uncertainty, Complex Morphology

Region	UAR	Boundary Ratio	Unc Inside	Unc Outside	LRP Corr
WT	0.490	0.642	0.00338	0.00005	−0.042
TC	**0.892**	0.776	0.01138	0.00001	−0.046
ET	0.624	**1.000**	0.00853	0.00003	−0.071

Table 7: MC Dropout uncertainty metrics for patient 01397. TC UAR of 0.892 indicates 89% of model uncertainty concentrates inside the Tumour Core. ET Boundary Ratio reaches 1.000, every unit of uncertainty sits at the ET edge.

UAR rises sharply to 0.538–0.907, with 91% of total uncertainty contained within the Tumour Core (UAR = 0.907). The ET boundary ratio reaches 0.998, localising virtually all uncertainty to the enhancing margin. Consistently negative LRP correlations (−0.050 to −0.071) indicate that regions of highest relevance coincide with lowest uncertainty, confirming the model remains confident in its most predictive features despite elevated overall ambiguity.

Patient 00291 – The Gradient-Based Failure Case

Region	UAR	Boundary Ratio	Unc Inside	Unc Outside	LRP Corr
WT	0.619	**1.000**	0.00440	0.00003	−0.012
TC	0.504	0.997	0.00802	0.00002	**+0.090**
ET	0.500	0.997	0.00930	0.00002	+0.053

Table 8: MC Dropout uncertainty metrics for patient 00291. Despite complete gradient-based XAI failure, Boundary Ratio reaches 0.997–1.000, proving the model's spatial reasoning is intact. The positive TC/ET correlation is unique to this patient.

This patient is uniquely diagnostic: Grad-CAM and Guided Grad-CAM produce **zero across all metrics** a complete gradient-based XAI failure. However, MC Dropout reveals a fundamentally different picture. The Boundary Ratio of 0.997–1.000 demonstrates that despite the model's ambiguity (UAR ≈ 0.50–0.62), uncertainty concentrates at the ground truth boundaries rather than being randomly distributed. This confirms the model *is* segmenting based on real spatial features—the gradient-based methods failed to explain it, but the model's internal reasoning remains spatially grounded.

The positive TC/ET Saliency-Uncertainty Correlation (+0.05 to +0.09) is unique to this patient. Where LRP assigns high relevance and MC Dropout assigns high uncertainty overlap slightly, suggesting the model relies on features it is not fully confident about. Clinically, this is a valuable flag: cases exhibiting this pattern should be prioritised for radiologist review.

Cross-Paradigm Finding: Saliency ≠ Uncertainty

Across all three patients and nine regional measurements, the Saliency-Uncertainty Correlation ranges from −0.071 to +0.090 with a mean near zero. This demonstrates that saliency (what the model attends to) and uncertainty (where the model doubts) are **independent, non-redundant signals**. This independence is a positive finding: if they were correlated, one signal would be redundant. Because they are independent, a clinician using this system receives two complementary tools—saliency maps for trust calibration ("Is the AI looking at the right features?") and uncertainty maps for risk assessment ("Where might the AI be wrong?"). This directly motivates deploying both modalities in the immersive VR environment presented in Pillar 3.

4.2.8 Cross-Method Comparative Analysis

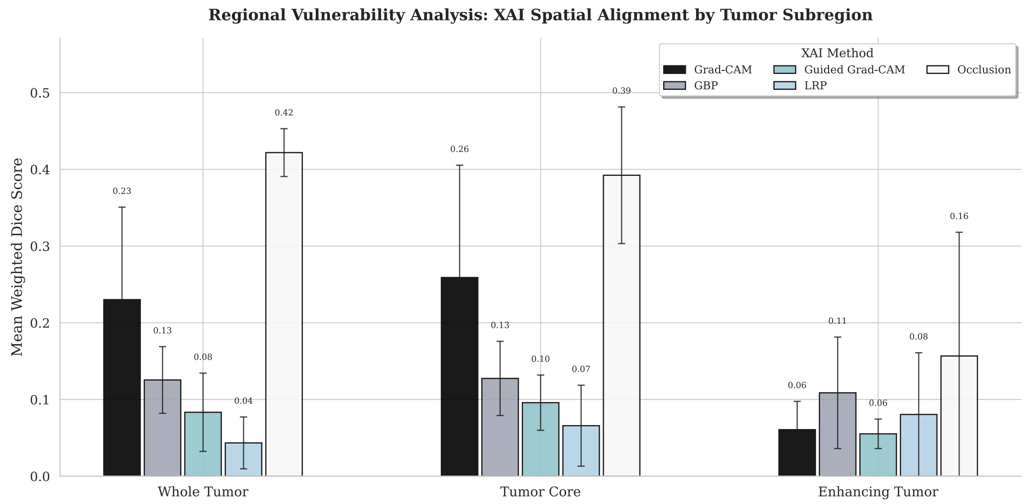


Figure 15: Regional Vulnerability Analysis: Mean Weighted Dice across five XAI attribution methods, grouped by tumour subregion. Occlusion Sensitivity dominates Whole Tumour and Tumour Core, while ET remains universally challenging. Error bars indicate standard deviation.

Method	WT Mean W.Dice	TC Mean W.Dice	ET Mean W.Dice
Grad-CAM	0.220	0.268	0.048
GBP	0.130	0.139	0.140
Guided Grad-CAM	0.133	0.153	0.224
LRP	0.043	0.058	0.071
Occlusion	**0.422**	**0.392**	**0.165**

Table 9: Mean Weighted Dice scores by tumour region across all five saliency-based XAI methods. Occlusion Sensitivity achieves the highest scores for WT and TC. Bold values indicate the best-performing method per region.

The Regional Vulnerability Analysis shows that Occlusion Sensitivity leads Whole Tumour (0.422) and Tumour Core (0.392), confirming its physically grounded model reliance. Enhancing Tumour remains the weakest region across all methods, revealing a fundamental limitation of interpretability techniques on small, heterogeneous subregions. GBP and Guided Grad-CAM outperform Grad-CAM on ET (0.14–0.22 versus 0.048), underscoring the advantage of full-voxel resolution.

A methodological insight from LRP further clarifies these rankings: despite recording the lowest Weighted Dice overall, LRP maintains 100% Pointing Game across every region. This confirms that Weighted Dice measures shape alignment while Pointing Game measures peak localisation, distinct and complementary dimensions of saliency quality. Neither metric alone is sufficient; together they provide a complete evaluation.

4.2.9 Visual Evaluation

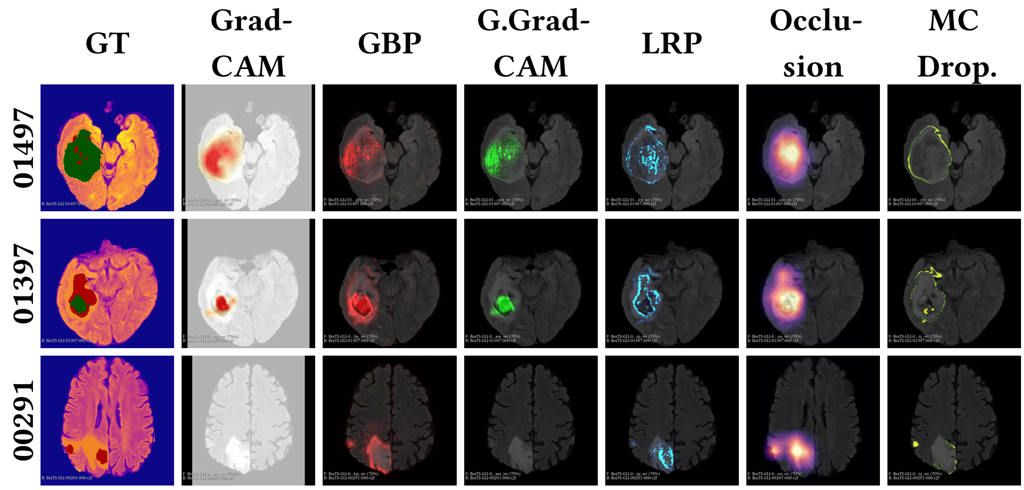


Figure 16: Multi-method XAI comparison across three patients with varying tumour morphologies. Patient 00291 (bottom row) demonstrates the gradient-based failure case: Grad-CAM and Guided Grad-CAM produce blank maps, while GBP, LRP, Occlusion, and MC Dropout continue to provide meaningful spatial information.

The visual grid provides an immediate, qualitative validation of the quantitative findings. Patient 01497 (top row) shows consistent, well-localised saliency across all methods, corresponding to its clear tumour boundaries and low MC Dropout uncertainty. Patient 01397 (middle row) exhibits more diffuse saliency patterns consistent with its complex morphology and high uncertainty scores. Patient 00291 (bottom row) visually confirms the gradient-based failure: the Grad-CAM and Guided Grad-CAM columns appear blank, while GBP, LRP, Occlusion, and MC Dropout continue to provide spatially meaningful explanations, visually corroborating the quantitative finding that perturbation based and uncertainty based methods are more robust to morphological edge cases.

4.3 Virtual Reality (VR) Immersive Visualisation

Having established clinically viable segmentation and multi-paradigm XAI validation, Pillar 3 demonstrates the complete end-to-end immersive pipeline and shows how it enables a clinician to interact with the model's spatial reasoning in real time.

4.3.1 Technical Implementation

The VR stack runs entirely on open, Linux-native tooling. A Bash launcher (`run_slicer_vr.sh`) sets `XR_RUNTIME_JSON` to the WiVRn OpenXR runtime manifest, directing wireless streaming to the Meta Quest 3, and `VTK_OPENXR_ACTION_MANIFEST` to the SlicerVR controller binding file, then launches 3D Slicer with `--python-script fix_vr.py`. This script executes five auto-initialisation functions at startup without any manual interaction: `enable_volume_rendering_for_all` activates GPU Ray Cast rendering (`vtkMRMLGPURayCastVolumeRenderingDisplayNode`) for every loaded NIfTI volume, preventing the CPU-bound fallback that causes session hangs; `setup_vr_view` creates the VR view node and activates the headset stream; `attach_custom_models` spawns geometry-accurate hand controllers bound to the left and right transform nodes; `update_vr_pointers` launches a 50 ms ray-march loop that tests voxel scalar values along each controller's forward axis, placing a green hit-dot at the first intersection; and `watch_for_new_volumes` polls every 5 seconds to auto-activate rendering whenever a new volume is loaded mid-session. The entire setup reduces a twelve-step manual workflow to a single terminal command.

4.3.2 Qualitative Evaluation: Brain and Isolated Tumour View

Figure 17 presents a side-by-side rendering full brain context alongside an isolated tumor-only view. The clinician can toggle between global anatomical context and subregion morphology, pointing directly at any voxel with the ray-marching controller cursor. This is a proof-of-concept instance, the pipeline is an active interrogation tool, not a passive display.

Figure 17: Side-by-side immersive rendering: full brain context (left) and isolated tumor view (right). The polychromatic subregion hierarchy is navigable from any orientation using 6-DoF hand controllers.

4.3.3 Qualitative Evaluation: XAI Heatmap Rendering in VR

Figure 18 renders the Occlusion Sensitivity heatmap for the same patient selected because it achieved the highest Weighted Dice scores across all six XAI methods (WT: 0.422, TC: 0.392). The `watch_for_new_volumes` watcher activates GPU rendering automatically when the saliency NIfTI is loaded. The warm attribution plume co-localises tightly with the segmentation boundary, closing the interpretability chain the clinician perceives in three dimensions which spatial regions drove the model's prediction, a verification structurally inaccessible in any flat-panel workflow.

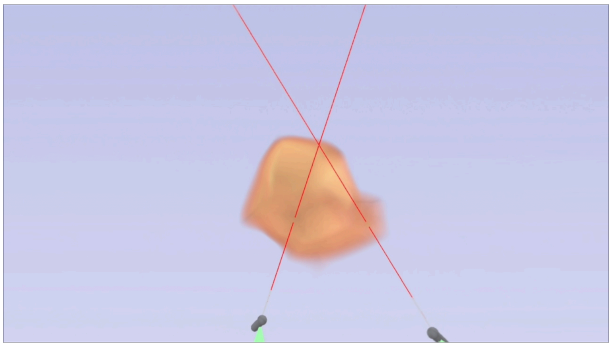


Figure 18: Occlusion Sensitivity heatmap rendered volumetrically in VR. Peak attribution (warm tones) envelops the tumor boundary, materialising the Weighted Dice localisation scores from Pillar 2 in three-dimensional perceptual space.

Chapter 5: Evaluation

Chapter 6: Conclusion

Chapter 7: Code

Use code blocks for snippets. Do not dump 1000s of lines here; use appendices for that. [cite: 20, 21]

java static public void main(String[] args) { try { UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName()); } catch(Exception e) { e.printStackTrace(); } new WelcomeApp(); }

Listing 1: Java Example

Chapter 8: Initial Project Proposal

Chapter 9: Final Project Proposal

Chapter 10: Application for Approval of Research Activities

Chapter 11: Client Consent Form

Chapter 11: Bibliography

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